

Error Signaling Module (ESM)

This chapter provides the details of the error signaling module (ESM) that aggregates device errors and provides the capability to define internal and external error response based on error severity.

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12.1 Overview

The Error Signaling Module (ESM) collects and reports the various error conditions on the microcontroller. The error condition is categorized based on a severity level. Error response is then generated based on the category of the error. Possible error responses include a low-priority interrupt, high-priority interrupt, and an external pin action.

12.1.1 Features

- Up to 128 error channels are supported, divided into 3 different groups:
 - 64 Group1 (low-severity) channels with configurable interrupt generation and configurable $\overline{\text{ERROR}}$ pin behavior
 - 32 Group2 (high-severity) channels with predefined interrupt generation and predefined $\overline{\text{ERROR}}$ pin behavior
 - 32 Group3 (high-severity) channels with no interrupt generation and predefined $\overline{\text{ERROR}}$ pin behavior. These channels have no interrupt response as they are reserved for CPU based diagnostics that generate aborts directly to the CPU.
- Dedicated device $\overline{\text{ERROR}}$ pin to signal an external observer
- Configurable timebase for $\overline{\text{ERROR}}$ pin output
- Error forcing capability for latent fault testing

12.1.2 Block Diagram

As shown in Figure 12-1, the ESM channels are divided into three groups. Group1 channels are considered to be low-severity. Group1 errors have a configurable interrupt response and configurable $\overline{\text{ERROR}}$ pin behavior. Note that the ESM Status Register 1 (ESMSR1) for error group1 gets updated, regardless of whether an ESM interrupt for that Group1 channel is enabled or not. Group2 channels are connected to higher-severity error signals. Group2 errors generate a non-maskable high-priority interrupt to the CPU and assert the $\overline{\text{ERROR}}$ pin. Group3 channels indicate errors of the highest severity. Check the specific part's datasheet for identifying group3 errors and their expected responses. Group3 errors always generate an $\overline{\text{ERROR}}$ pin output.

The ESM interrupt and $\overline{\text{ERROR}}$ pin behavior are also summarized in Table 12-1.

Figure 12-1. Block Diagram

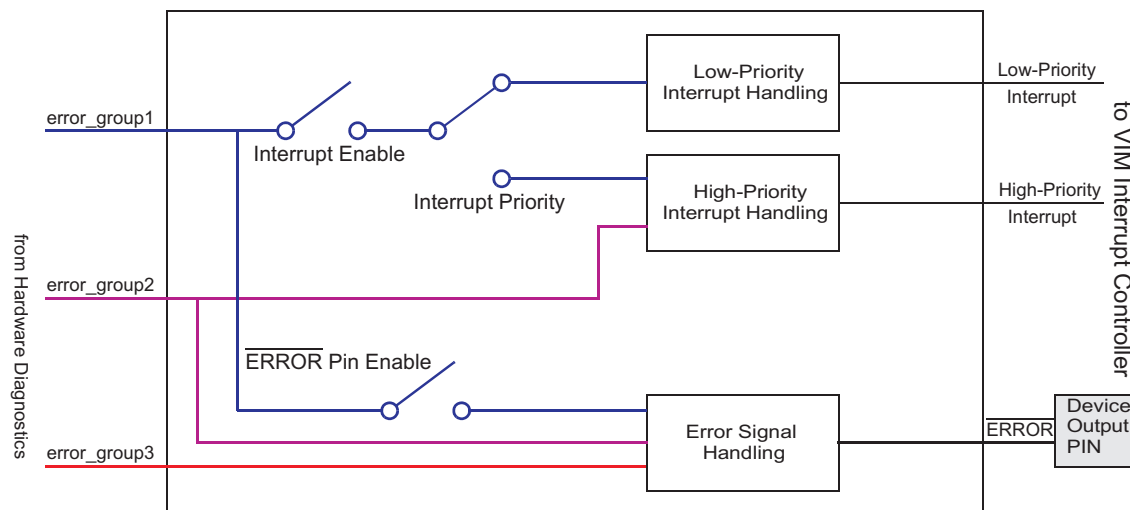


Table 12-1. ESM Interrupt and $\overline{\text{ERROR}}$ Pin Behavior

Error Group	Interrupt to CPU	Interrupt Priority	$\overline{\text{ERROR}}$ Pin Response
1	Can be enabled or disabled for each channel	Can be selected as low/high-priority for each channel	$\overline{\text{ERROR}}$ pin action can be selected for each channel separately
2	Cannot be disabled	High priority	$\overline{\text{ERROR}}$ pin is asserted
3	No interrupt	NA	$\overline{\text{ERROR}}$ pin is asserted

Figure 12-2 and Figure 12-3 show the interrupt response handling and $\overline{\text{ERROR}}$ pin response handling with register configuration. The total active time of the $\overline{\text{ERROR}}$ pin is controlled by the Low-Time Counter Preload register (LTCP) and the key register (ESMEPSR) as shown in Figure 12-3. See Section 12.2.2 for details.

Figure 12-2. Interrupt Response Handling

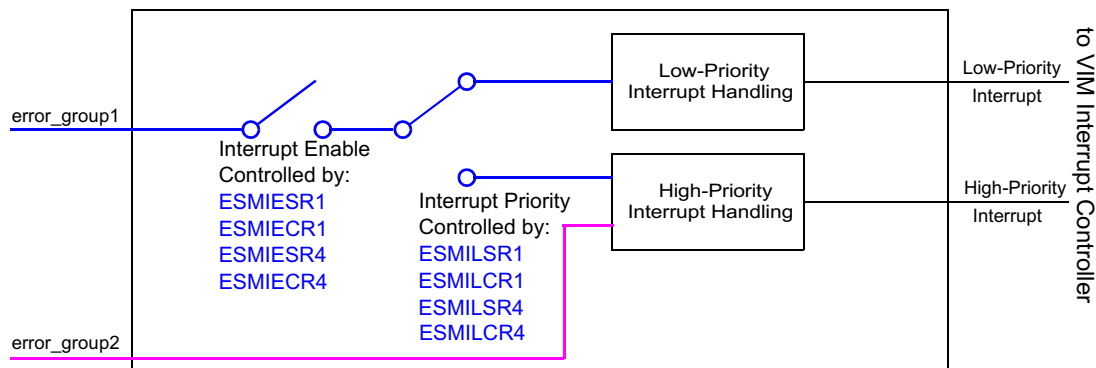
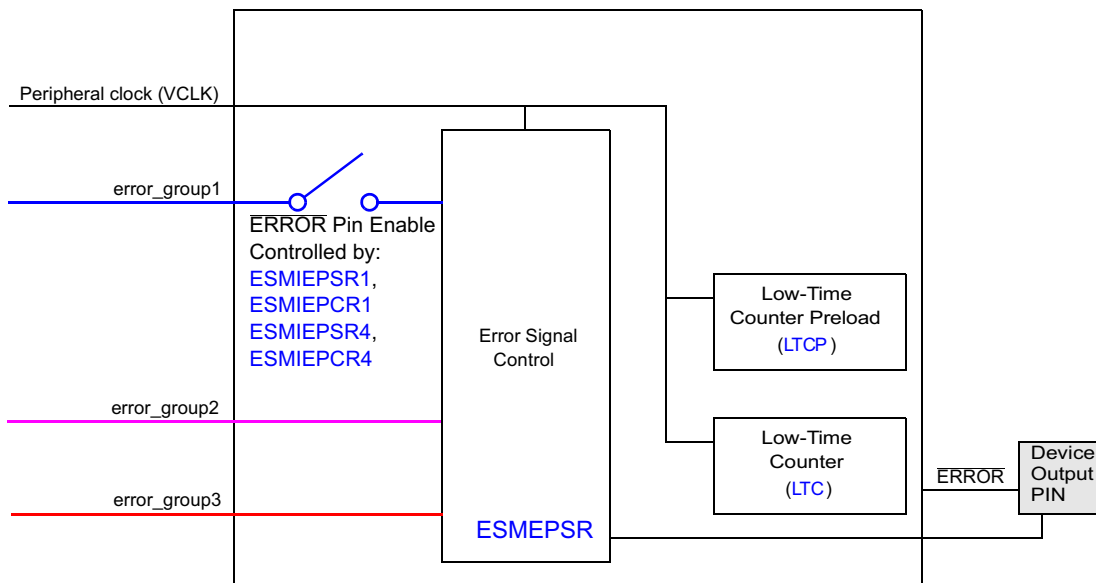


Figure 12-3. $\overline{\text{ERROR}}$ Pin Response Handling



12.2 Module Operation

This device has 128 error channels, divided into 3 different error groups. Please refer to the device datasheet for ESM channel assignment details.

The ESM module has error flags for each error channel. The error status registers ESMSR1, ESMSR4, ESMSR2, ESMSR3 provide status information on a pending error of Group1 (Channel 0-31), Group1 (Channel 32-63), Group2, and Group3, respectively. The ESMEPSR register provides the current $\overline{\text{ERROR}}$ status. The module also provides a status shadow register, ESMSSR2, which maintains the error flags of Group2 until power-on reset ($\overline{\text{PORRST}}$) is asserted. See [Section 12.2.1](#) for details of their behavior during power on reset and warm reset.

Once an error occurs, the ESM module will set the corresponding error flags. In addition, it can trigger an interrupt, $\overline{\text{ERROR}}$ pin outputs low depending on the ESM settings. Once the $\overline{\text{ERROR}}$ pin outputs low, a power on reset or a write of 5h to ESMEKR is required to release the ESM error pin back to normal state. See [Section 12.2.2](#) for details. The application can read the error status registers (ESMSR1, ESMSR4, ESMSR2, and ESMSR3) to debug the error. If an $\overline{\text{RST}}$ is triggered or the error interrupt has been served, the error flag of Group2 should be read from ESMSSR2 because the error flag in ESMSR2 will be cleared by $\overline{\text{RST}}$.

The user can also test the functionality of the $\overline{\text{ERROR}}$ pin by forcing an error. See [Section 12.2.3](#) for details.

12.2.1 Reset Behavior

Power on reset:

- $\overline{\text{ERROR}}$ pin behavior

When $\overline{\text{PORRST}}$ is active, the $\overline{\text{ERROR}}$ pin is in a high impedance state (output drivers disabled).

- Register behavior

After $\overline{\text{PORRST}}$, all registers in ESM module will be re-initialized to the default value. All the error status registers are cleared to zero.

Warm reset ($\overline{\text{RST}}$):

- $\overline{\text{ERROR}}$ pin behavior

During $\overline{\text{RST}}$, the $\overline{\text{ERROR}}$ pin is in “output active” state with pull-down disabled. The $\overline{\text{ERROR}}$ pin remains unchanged after $\overline{\text{RST}}$.

- Register behavior

After $\overline{\text{RST}}$, ESMSR1, ESMSR4, ESMSSR2, ESMSR3 and ESMEPSR register values remains unchanged. Since $\overline{\text{RST}}$ does not clear the critical failure registers, the user can read those registers to debug the failures after $\overline{\text{RST}}$ pin goes back to high.

After $\overline{\text{RST}}$, if one of the flags in ESMSR1 and ESMSR4 is set, the interrupt service routine will be called once the corresponding interrupt is enabled.

NOTE: ESMSR2 is cleared after $\overline{\text{RST}}$. The flag in ESMSR2 gets cleared when reading the appropriate vector in the ESMIOFFHR offset register. Reading ESMIOFFHR will not clear the ESMSR1, ESMSR4, and the shadow register ESMSSR2. Reading ESMIOFFLR will also not clear the ESMSR1 and ESMSR4.

12.2.2 $\overline{\text{ERROR}}$ Pin Timing

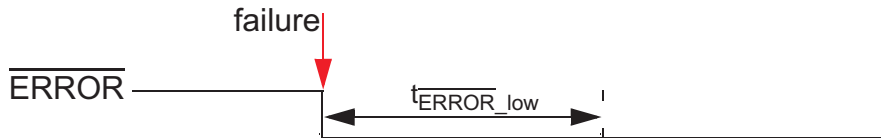
The $\overline{\text{ERROR}}$ pin is an active-low function. The state of the pin is also readable from $\overline{\text{ERROR}}$ Pin Status Register (ESMEPSR). The pin is in a high-impedance state during power-on reset. Once the ESM module drives the $\overline{\text{ERROR}}$ pin low, it remains in this state for the time specified by the Low-Time Counter Preload register (LTCPR). Based on the time period of the peripheral clock (VCLK), the total active time of the $\overline{\text{ERROR}}$ pin can be calculated as:

$$t_{\overline{\text{ERROR_low}}} = t_{\text{VCLK}} \times (\text{LTCP} + 1) \quad (22)$$

Once this period expires, the $\overline{\text{ERROR}}$ pin is set to high in case the reset of the $\overline{\text{ERROR}}$ pin was requested. This request is done by writing an appropriate key (5h) to the key register (ESMEKR) during the $\overline{\text{ERROR}}$ pin low time. Here are a few examples:

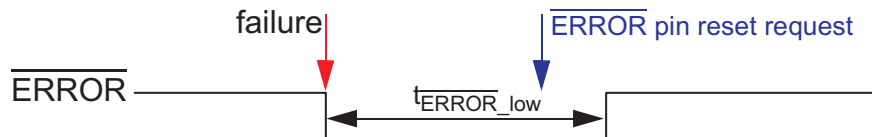
Example 1: ESM detects a failure and drives the $\overline{\text{ERROR}}$ pin low. No $\overline{\text{ERROR}}$ pin reset is requested. The $\overline{\text{ERROR}}$ pin continues outputting low until power on reset occurs.

Figure 12-4. $\overline{\text{ERROR}}$ Pin Timing - Example 1



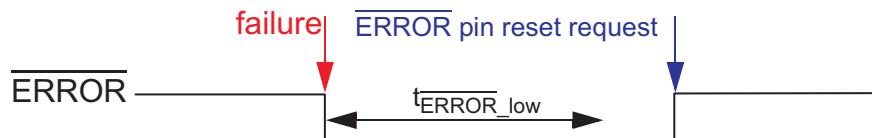
Example 2: ESM detects a failure and drives the $\overline{\text{ERROR}}$ pin low. An $\overline{\text{ERROR}}$ pin reset request is received before $t_{\overline{\text{ERROR_low}}}$ expires. In this case, the $\overline{\text{ERROR}}$ pin is set to high immediately after $t_{\overline{\text{ERROR_low}}}$ expires.

Figure 12-5. $\overline{\text{ERROR}}$ Pin Timing - Example 2



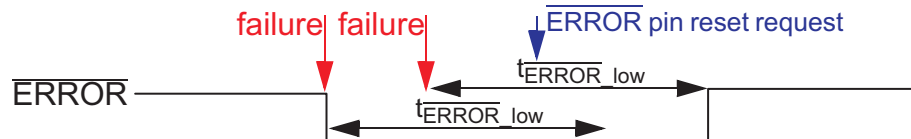
Example 3: ESM detects a failure and drives the $\overline{\text{ERROR}}$ pin low. An $\overline{\text{ERROR}}$ pin reset request is received after $t_{\overline{\text{ERROR_low}}}$ expires. In this case, the $\overline{\text{ERROR}}$ pin is set to high immediately after $\overline{\text{ERROR}}$ pin reset request is received.

Figure 12-6. $\overline{\text{ERROR}}$ Pin Timing - Example 3



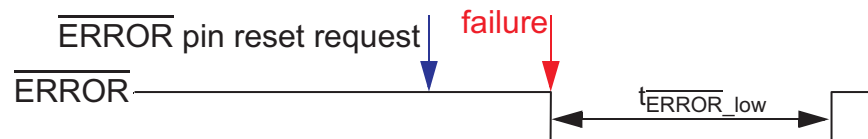
Example 4: ESM detects a failure and drives the $\overline{\text{ERROR}}$ pin low. Another failure occurs within the time the pin stays low. In this case, the low-time counter will be reset when the other failure occurs. In other words, $t_{\overline{\text{ERROR}}_{\text{low}}}$ should be counted from whenever the most recent failure occurs.

Figure 12-7. $\overline{\text{ERROR}}$ Pin Timing - Example 4



Example 5: The reset of the $\overline{\text{ERROR}}$ pin was requested by the software even before the failure occurs. In this case, the $\overline{\text{ERROR}}$ pin is set to high immediately after $t_{\overline{\text{ERROR}}_{\text{low}}}$ expires. This case is not recommended and should be avoided by the application.

Figure 12-8. $\overline{\text{ERROR}}$ Pin Timing - Example 5



Example 6: Failure1, then $\overline{\text{ERROR}}$ pin reset request, then Failure2 occurs before $\overline{\text{ERROR}}$ pin gets reset. In this case, the $\overline{\text{ERROR}}$ pin low-time is just extended (restarted) when the failure2 occurs and goes high when this count-down expires. There now is a scenario where the $\overline{\text{ERROR}}$ pin is high and the group2/3 status flag is set. To avoid this scenario, the application must write 5h followed by 0 to the ESM Error Key Register (ESMEKR). In this case, the $\overline{\text{ERROR}}$ pin will go high and then go low again to indicate the second failure.

12.2.3 Forcing an Error Condition

The error response generation mechanism is testable by software by forcing an error condition. This allows testing the $\overline{\text{ERROR}}$ pin functionality. By writing a dedicated key to the ESM Error Key Register (ESMEKR), the $\overline{\text{ERROR}}$ pin is set to low for the specified time. The following steps describe how to force an error condition:

1. Check $\overline{\text{ERROR}}$ Pin Status Register (ESMEPSR). This register must be 1 to switch into the error forcing mode.

The ESM module cannot be switched into the error forcing mode if a failure has already been detected in functional mode. The application command to switch to error forcing mode is ignored.

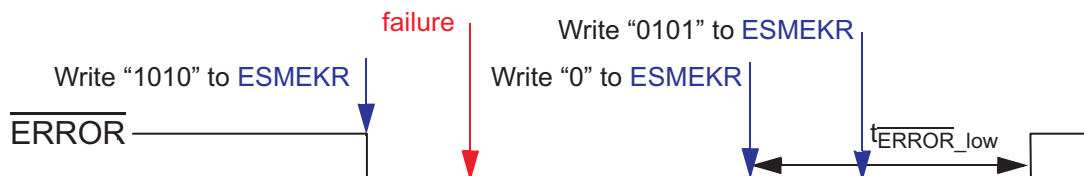
2. Write 5h to the ESM Error Key Register (ESMEKR). After that, the $\overline{\text{ERROR}}$ pin should output low (error force mode).

Once the application puts the ESM module in the error forcing mode, the $\overline{\text{ERROR}}$ pin cannot indicate the normal error functionality. If a failure occurs during this time, it gets still latched and the LTC is reset and stopped. The error output pin is already driven low on account of the error forcing mode. When the ESM is forced back to normal functional mode, the LTC becomes active and forces the $\overline{\text{ERROR}}$ pin low until the expiration of the LTC (see Figure 12-9).

3. Write 0 to the ESM Error Key Register (ESMEKR) back to the active normal mode.

If there are no errors detected while the ESM module is in the error forcing mode, the $\overline{\text{ERROR}}$ pin goes high immediately after exiting the error forcing mode.

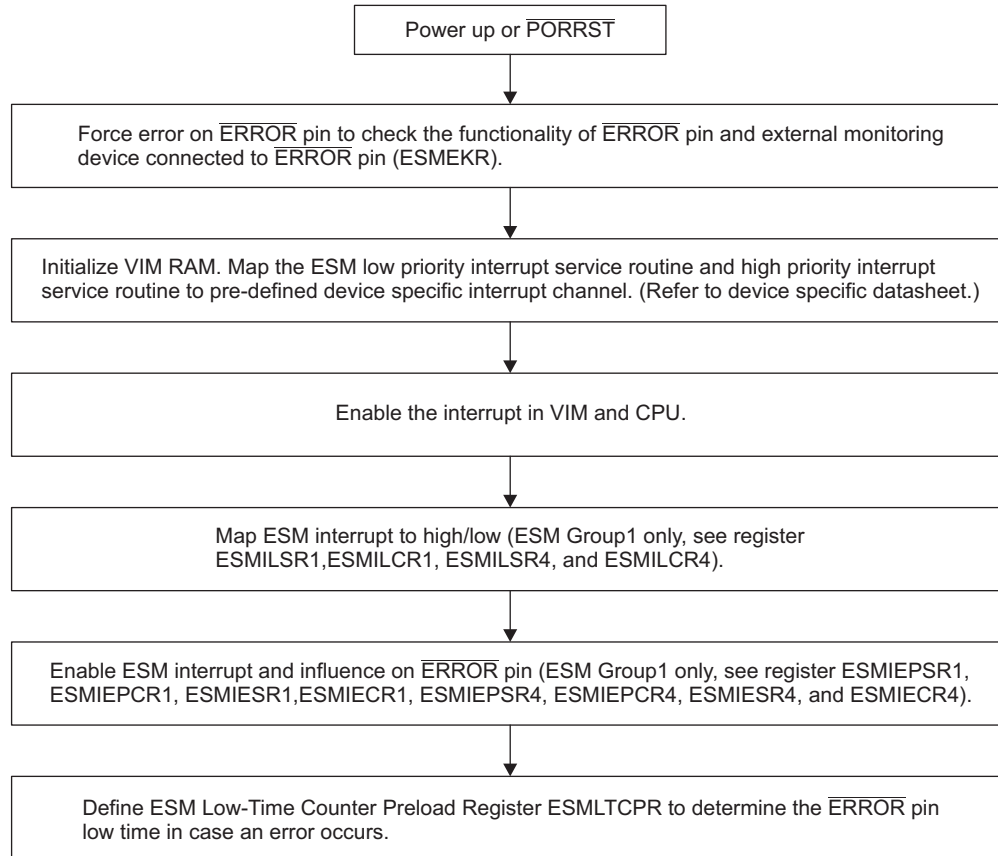
Figure 12-9. $\overline{\text{ERROR}}$ Pin Timing - Example 7



12.3 Recommended Programming Procedure

During the initialization stage, the application code should follow the recommendations in [Figure 12-10](#) to initialize the ESM.

Figure 12-10. ESM Initialization



Once an error occurs, it can trigger an interrupt, $\overline{\text{ERROR}}$ pin outputs low depending on the ESM settings. Once the $\overline{\text{ERROR}}$ pin outputs low, a power on reset or a write of 5h to ESMEKR is required to release the ESM back to normal state. The application can read the error status registers (ESMSR1, ESMSR4, ESMSR2, and ESMSR3) to debug the error. If an $\overline{\text{RST}}$ is triggered or the error interrupt has been served, the error flag of Group2 should be read from ESMSR2 because the error flag in ESMSR2 will be cleared by $\overline{\text{RST}}$.

12.4 Control Registers

This section describes the ESM registers. Each register begins on a 32-bit word boundary. The registers support 32-bit, 16-bit, and 8-bit accesses. The base address for the registers is FFFF F500h.

Table 12-2. ESM Module Registers

Address	Acronym	Register Description	Section
FFFF F500h	ESMEEPAPR1	ESM Enable $\overline{\text{ERROR}}$ Pin Action/Response Register 1	Section 12.4.1
FFFF F504h	ESMDEPAPR1	ESM Disable $\overline{\text{ERROR}}$ Pin Action/Response Register 1	Section 12.4.2
FFFF F508h	ESMIESR1	ESM Interrupt Enable Set Register 1	Section 12.4.3
FFFF F50Ch	ESMIECR1	ESM Interrupt Enable Clear Register 1	Section 12.4.4
FFFF F510h	ESMILSR1	Interrupt Level Set Register 1	Section 12.4.5
FFFF F514h	ESMILCR1	Interrupt Level Clear Register 1	Section 12.4.6
FFFF F518h	ESMSR1	ESM Status Register 1	Section 12.4.7
FFFF F51Ch	ESMSR2	ESM Status Register 2	Section 12.4.8
FFFF F520h	ESMSR3	ESM Status Register 3	Section 12.4.9
FFFF F524h	ESMEPSR	ESM $\overline{\text{ERROR}}$ Pin Status Register	Section 12.4.10
FFFF F528h	ESMIOFFHR	ESM Interrupt Offset High Register	Section 12.4.11
FFFF F52Ch	ESMIOFFLR	ESM Interrupt Offset Low Register	Section 12.4.12
FFFF F530h	ESMLTCR	ESM Low-Time Counter Register	Section 12.4.13
FFFF F534h	ESMLTCPR	ESM Low-Time Counter Preload Register	Section 12.4.14
FFFF F538h	ESMEKR	ESM Error Key Register	Section 12.4.15
FFFF F53Ch	ESMSSR2	ESM Status Shadow Register 2	Section 12.4.16
FFFF F540h	ESMIEPSR4	ESM Influence $\overline{\text{ERROR}}$ Pin Set Register 4	Section 12.4.17
FFFF F544h	ESMIEPCR4	ESM Influence $\overline{\text{ERROR}}$ Pin Clear Register 4	Section 12.4.18
FFFF F548h	ESMIESR4	ESM Interrupt Enable Set Register 4	Section 12.4.19
FFFF F54Ch	ESMIECR4	ESM Interrupt Enable Clear Register 4	Section 12.4.20
FFFF F550h	ESMILSR4	Interrupt Level Set Register 4	Section 12.4.21
FFFF F554h	ESMILCR4	Interrupt Level Clear Register 4	Section 12.4.22
FFFF F558h	ESMSR4	ESM Status Register 4	Section 12.4.23

12.4.1 ESM Enable $\overline{\text{ERROR}}$ Pin Action/Response Register 1 (ESMEEPAPR1)

This register is dedicated for Group1.

Figure 12-11. ESM Enable $\overline{\text{ERROR}}$ Pin Action/Response Register 1 (ESMEEPAPR1)
[address = FFFF F500h]

31	16
IEPSET	
R/WP-0	
15	0
IEPSET	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 12-3. ESM Enable $\overline{\text{ERROR}}$ Pin Action/Response Register 1 (ESMEEPAPR1)
Field Descriptions

Bit	Field	Value	Description
31-0	IEPSET	0	Enable $\overline{\text{ERROR}}$ Pin Action/Response on Group 1. Read in User and Privileged mode. Write in Privileged mode only. Read: Failure on channel x has no influence on $\overline{\text{ERROR}}$ pin. Write: Leaves the bit and the corresponding clear bit in the ESMIEPCR1 register unchanged.
		1	Read: Failure on channel x has influence on $\overline{\text{ERROR}}$ pin. Write: Enables failure influence on $\overline{\text{ERROR}}$ pin and sets the corresponding clear bit in the ESMIEPCR1 register.

12.4.2 ESM Disable $\overline{\text{ERROR}}$ Pin Action/Response Register 1 (ESMDEPAPR1)

This register is dedicated for Group1.

Figure 12-12. ESM Disable $\overline{\text{ERROR}}$ Pin Action/Response Register 1 (ESMDEPAPR1)
[address = FFFF F504h]

31	16
IEPCLR	
R/WP-0	
15	0
IEPCLR	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 12-4. ESM Disable $\overline{\text{ERROR}}$ Pin Action/Response Register 1 (ESMDEPAPR1)
Field Descriptions

Bit	Field	Value	Description
31-0	IEPCLR	0	Disable $\overline{\text{ERROR}}$ Pin Action/Response on Group 1. Read in User and Privileged mode. Write in Privileged mode only. Read: Failure on channel x has no influence on $\overline{\text{ERROR}}$ pin. Write: Leaves the bit and the corresponding set bit in the ESMIEPSR1 register unchanged.
		1	Read: Failure on channel x has influence on $\overline{\text{ERROR}}$ pin. Write: Disables failure influence on $\overline{\text{ERROR}}$ pin and clears the corresponding set bit in the ESMIEPSR1 register.

12.4.3 ESM Interrupt Enable Set Register 1 (ESMIESR1)

This register is dedicated for Group1.

Figure 12-13. ESM Interrupt Enable Set Register 1 (ESMIESR1)
[address = FFFF F508h]

31	16
INTENSET	
R/WP-0	
15	0
INTENSET	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 12-5. ESM Interrupt Enable Set Register 1 (ESMIESR1) Field Descriptions

Bit	Field	Value	Description
31-0	INTENSET	0	Set interrupt enable. Read in User and Privileged mode. Write in Privileged mode only. Read: Interrupt is disabled. Write: Leaves the bit and the corresponding clear bit in the ESMIECR1 register unchanged.
		1	Read: Interrupt is enabled. Write: Enables interrupt and sets the corresponding clear bit in the ESMIECR1 register.

12.4.4 ESM Interrupt Enable Clear Register 1 (ESMIECR1)

This register is dedicated for Group1.

Figure 12-14. ESM Interrupt Enable Clear Register 1 (ESMIECR1)
[address = FFFF F50Ch]

31	16
INTENCLR	
R/WP-0	
15	0
INTENCLR	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 12-6. ESM Interrupt Enable Clear Register 1 (ESMIECR1) Field Descriptions

Bit	Field	Value	Description
31-0	INTENCLR	0	Clear interrupt enable. Read in User and Privileged mode. Write in Privileged mode only. Read: Interrupt is disabled. Write: Leaves the bit and the corresponding set bit in the ESMIESR1 register unchanged.
		1	Read: Interrupt is enabled. Write: Disables interrupt and clears the corresponding set bit in the ESMIESR1 register.

12.4.5 ESM Interrupt Level Set Register 1 (ESMILSR1)

This register is dedicated for Group1.

Figure 12-15. ESM Interrupt Level Set Register 1 (ESMILSR1)
[address = FFFF F510h]

31	INTLVLSET	16
	R/WP-0	
15	INTLVLSET	0
	R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 12-7. ESM Interrupt Level Set Register 1 (ESMILSR1) Field Descriptions

Bit	Field	Value	Description
31-0	INTLVLSET	0	Set interrupt priority. Read in User and Privileged mode. Write in Privileged mode only. Read: Interrupt of channel x is mapped to low-level interrupt line. Write: Leaves the bit and the corresponding clear bit in the ESMILCR1 register unchanged.
		1	Read: Interrupt of channel x is mapped to high-level interrupt line. Write: Maps interrupt of channel x to high-level interrupt line and sets the corresponding clear bit in the ESMILCR1 register.

12.4.6 ESM Interrupt Level Clear Register 1 (ESMILCR1)

This register is dedicated for Group1.

Figure 12-16. ESM Interrupt Level Clear Register 1 (ESMILCR1)
[address = FFFF F514h]

31	INTLVLCCLR	16
	R/WP-0	
15	INTLVLCCLR	0
	R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 12-8. ESM Interrupt Level Clear Register 1 (ESMILCR1) Field Descriptions

Bit	Field	Value	Description
31-0	INTLVLCCLR	0	Clear interrupt priority. Read in User and Privileged mode. Write in Privileged mode only. Read: Interrupt of channel x is mapped to low-level interrupt line. Write: Leaves the bit and the corresponding set bit in the ESMILSR1 register unchanged.
		1	Read: Interrupt of channel x is mapped to high-level interrupt line. Write: Maps interrupt of channel x to low-level interrupt line and clears the corresponding set bit in the ESMILSR1 register.

12.4.7 ESM Status Register 1 (ESMSR1)

This register is dedicated for Group1. Note that the ESMSR1 status register will get updated if an error condition occurs, regardless if the corresponding interrupt enable flag is set or not.

Figure 12-17. ESM Status Register 1 (ESMSR1)
[address = FFFF F518h]

31	16
ESF	
R/W1CP-X/0	
15	0
ESF	
R/W1CP-X/0	

LEGEND: R/W = Read/Write; W1CP = Write 1 to clear in privilege mode only; -n = value after reset/PORRST; X = Value unchanged

Table 12-9. ESM Status Register 1 (ESMSR1) Field Descriptions

Bit	Field	Value	Description
31-0	ESF	0	Error Status Flag. Provides status information on a pending error. Read in User and Privileged mode. Write in Privileged mode only. Read: No error occurred; no interrupt is pending. Write: Leaves the bit unchanged.
		1	Read: Error occurred; interrupt is pending. Write: Clears the bit. Note: After $\overline{RST}$, if one of these flags are set and the corresponding interrupt are enabled, the interrupt service routine will be called.

12.4.8 ESM Status Register 2 (ESMSR2)

This register is dedicated for Group2.

Figure 12-18. ESM Status Register 2 (ESMSR2)
[address = FFFF F51Ch]

31	16
ESF	
R/W1CP-0	
15	0
ESF	
R/W1CP-0	

LEGEND: R/W = Read/Write; W1CP = Write 1 to clear in privilege mode only; -n = value after reset

Table 12-10. ESM Status Register 2 (ESMSR2) Field Descriptions

Bit	Field	Value	Description
31-0	ESF	0	Error Status Flag. Provides status information on a pending error. Read in User and Privileged mode. Write in Privileged mode only. Read: No error occurred; no interrupt is pending. Write: Leaves the bit unchanged.
		1	Read: Error occurred; interrupt is pending. Write: Clears the bit. ESMSR2 is not impacted by this action. Note: In normal operation, the flag gets cleared when reading the appropriate vector in the ESMIOFFHR offset register. Reading ESMIOFFHR will not clear the ESMSR1 and the shadow register ESMSR2.

12.4.9 ESM Status Register 3 (ESMSR3)

This register is dedicated for Group3.

Figure 12-19. ESM Status Register 3 (ESMSR3)
[address = FFFF F520h]

31	16
ESF	
R/W1CP-X/0	
15	0
ESF	
R/W1CP-X/0	

LEGEND: R/W = Read/Write; W1CP = Write 1 to clear in privilege mode only; -n = value after reset/PORRST; X = Value unchanged

Table 12-11. ESM Status Register 3 (ESMSR3) Field Descriptions

Bit	Field	Value	Description
31-0	ESF	0	Error Status Flag. Provides status information on a pending error. Read in User and Privileged mode. Write in Privileged mode only. Read: No error occurred. Write: Leaves the bit unchanged.
		1	Read: Error occurred. Write: Clears the bit.

12.4.10 ESM $\overline{\text{ERROR}}$ Pin Status Register (ESMEPSR)

Figure 12-20. ESM $\overline{\text{ERROR}}$ Pin Status Register (ESMEPSR)
[address = FFFF F524h]

31	16
Reserved	
R-0	
15	1 0
Reserved	
R-0	
EPSF	
R-X/1	

LEGEND: R = Read only; -n = value after reset/PORRST; X = Value unchanged

Table 12-12. ESM $\overline{\text{ERROR}}$ Pin Status Register (ESMEPSR) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	EPSF	0	$\overline{\text{ERROR}}$ Pin Status Flag. Provides status information for the $\overline{\text{ERROR}}$ pin. Read/Write in User and Privileged mode. Read: $\overline{\text{ERROR}}$ pin is low (active) if any error has occurred. Write: Writes have no effect.
		1	Read: $\overline{\text{ERROR}}$ pin is high if no error has occurred. Write: Writes have no effect. Note: This flag will be set to 1 after PORRST. The value will be unchanged after RST. The $\overline{\text{ERROR}}$ pin status remains unchanged after RST.

12.4.11 ESM Interrupt Offset High Register (ESMIOFFHR)

Figure 12-21. ESM Interrupt Offset High Register (ESMIOFFHR)
[address = FFFF F528h]

31	Reserved										16	
R-0												
15	Reserved					7	6	INTOFFH				0
R-0							R-0					

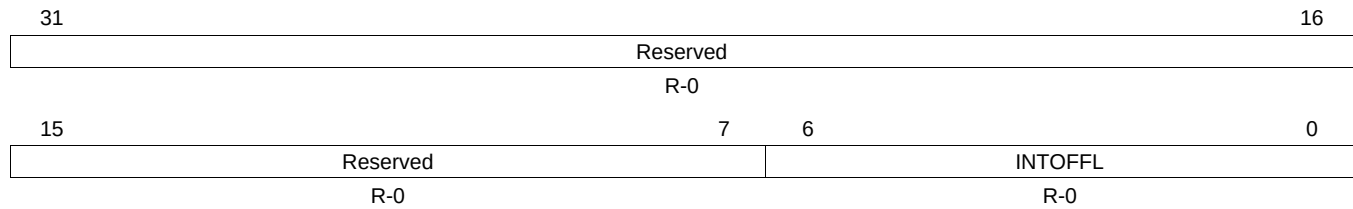
LEGEND: R = Read only; -n = value after reset

Table 12-13. ESM Interrupt Offset High Register (ESMIOFFHR) Field Descriptions

Bit	Field	Value	Description
31-7	Reserved	0	Reads return 0. Writes have no effect.
6-0	INTOFFH	Offset High-Level Interrupt. This vector gives the channel number of the highest-pending interrupt request for the high-level interrupt line. Interrupts of error Group2 have higher priority than interrupts of error Group1. Inside a group, channel 0 has highest priority and channel 31 has lowest priority. User and privileged mode (read): Returns number of pending interrupt with the highest priority for the high-level interrupt line. 0 No pending interrupt. 1h Interrupt pending for channel 0, error Group1. : 20h Interrupt pending for channel 31, error Group1. 21h Interrupt pending for channel 0, error Group2. : 40h Interrupt pending for channel 31, error Group2. 41h Interrupt pending for channel 32, error Group1. : 60h Interrupt pending for channel 63, error Group1. Note: Reading the interrupt vector will clear the corresponding flag in the ESMSR2 register; will not clear ESMSR1 and ESMSSR2 and the offset register gets updated. User and privileged mode (write): Writes have no effect.	

12.4.12 ESM Interrupt Offset Low Register (ESMIOFFLR)

Figure 12-22. ESM Interrupt Offset Low Register (ESMIOFFLR)
[address = FFFF F52Ch]



LEGEND: R = Read only; -n = value after reset

Table 12-14. ESM Interrupt Offset Low Register (ESMIOFFLR) Field Descriptions

Bit	Field	Value	Description
31-7	Reserved	0	Reads return 0. Writes have no effect.
6-0	INTOFFL	Offset Low-Level Interrupt. This vector gives the channel number of the highest-pending interrupt request for the low-level interrupt line. Inside a group, channel 0 has highest priority and channel 31 has lowest priority. User and privileged mode (read): Returns number of pending interrupt with the highest priority for the low-level interrupt line. 0 No pending interrupt. 1h Interrupt pending for channel 0, error Group1. : : 20h Interrupt pending for channel 31, error Group1. 21h-40h Reserved 41h Interrupt pending for channel 32, error Group1. : : 60h Interrupt pending for channel 63, error Group1. Note: Reading the interrupt vector will not clear the corresponding flag in the ESMSR1 register. Group2 interrupts are fixed to the high-level interrupt line only.	User and privileged mode (write): Writes have no effect.

12.4.13 ESM Low-Time Counter Register (ESMLTCR)

Figure 12-23. ESM Low-Time Counter Register (ESMLTCR)
[address = FFFF F530h]

31	Reserved	16
R-0		
15	LTC	0
R-3FFFh		

LEGEND: R = Read only; -n = value after reset

Table 12-15. ESM Low-Time Counter Register (ESMLTCR) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	LTC		ERROR Pin Low-Time Counter 16-bit preloadable down-counter to control low-time of ERROR pin. The low-time counter is triggered by the peripheral clock (VCLK). Note: Low-time counter is set to the default preload value of the ESMLTCPR in the following cases: 1. Reset (power on reset or warm reset) 2. An error occurs 3. User forces an error

12.4.14 ESM Low-Time Counter Preload Register (ESMLTCPR)

Figure 12-24. ESM Low-Time Counter Preload Register (ESMLTCPR)
[address = FFFF F534h]

31	Reserved	16
R-0		
15	LTCP	0
14	LTCP	
13	LTCP	
R/WP-0 R-3FFFh		

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 12-16. ESM Low-Time Counter Preload Register (ESMLTCPR) Field Descriptions

Bit	Field	Value	Description
31-16	Reserved	0	Reads return 0. Writes have no effect.
15-0	LTCP	0-FFFFh	ERROR Pin Low-Time Counter Pre-load Value 16-bit preload value for the ERROR pin low-time counter. Defines the minimum period for which the ERROR pin will be driven to 16384 VCLK cycles. Note: Only LTCP[15] and LTCP[14] are configurable (privileged mode write).

12.4.15 ESM Error Key Register (ESMEKR)

Figure 12-25. ESM Error Key Register (ESMEKR)
[address = FFFF F538h]

31	Reserved																16								
R-0																									
15	Reserved												4	3	EKEY										0
R-0														R/WP-0											

LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 12-17. ESM Error Key Register (ESMEKR) Field Descriptions

Bit	Field	Value	Description
31-4	Reserved	0	Reads return 0. Writes have no effect.
3-0	EKEY		Error Key. The key to reset the ERROR pin or to force an error on the ERROR pin. User and privileged mode (read): Returns current value of the EKEY.
		0	Privileged mode (write): Activates normal mode (recommended default mode).
		5h	The ERROR pin set to high when the low-time counter (LTC) has completed; then the EKEY bit will switch back to normal mode (EKEY = 0000).
		Ah	Forces error on ERROR pin.
		All other values	Activates normal mode.

12.4.16 ESM Status Shadow Register 2 (ESMSSR2)

This register is dedicated for Group2.

Figure 12-26. ESM Status Shadow Register 2 (ESMSSR2)
[address = FFFF F53Ch]

31																	16
ESF																	
R/W1CP-X/0																	
15																	0
ESF																	
R/W1CP-X/0																	

LEGEND: R/W = Read/Write; W1CP = Write 1 to clear in privilege mode only; -n = value after reset/**PORRST**; X = Value unchanged

Table 12-18. ESM Status Shadow Register 2 (ESMSSR2) Field Descriptions

Bit	Field	Value	Description
31-0	ESF		Error Status Flag. Shadow register for status information on pending error. Read in User and Privileged mode. Write in Privileged mode only.
		0	Read: No error occurred. Write: Leaves the bit unchanged.
		1	Read: Error occurred. Write: Clears the bit. ESMSSR2 is not impacted by this action. Note: Errors are stored until they are cleared by the software or at power-on reset (PORRST).

12.4.17 ESM Influence $\overline{\text{ERROR}}$ Pin Set Register 4 (ESMIEPSR4)

This register is dedicated for Group1.

Figure 12-27. ESM Influence $\overline{\text{ERROR}}$ Pin Set Register 4 (ESMIEPSR4)
[address = FFFF F540h]

31	16
IEPSET[63:48]	
R/WP-0	
15	0
IEPSET[47:32]	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 12-19. ESM Influence $\overline{\text{ERROR}}$ Pin Set Register 4 (ESMIEPSR4) Field Descriptions

Bit	Field	Value	Description
63-32	IEPSET	0	Set influence on $\overline{\text{ERROR}}$ pin. Read in User and Privileged mode. Write in Privileged mode only. Read: Failure on channel x has no influence on $\overline{\text{ERROR}}$ pin. Write: Leaves the bit and the corresponding clear bit in the ESMIEPCR4 register unchanged.
		1	Read: Failure on channel x has influence on $\overline{\text{ERROR}}$ pin. Write: Enables failure influence on $\overline{\text{ERROR}}$ pin and sets the corresponding clear bit in the ESMIEPCR4 register.

12.4.18 ESM Influence $\overline{\text{ERROR}}$ Pin Clear Register 4 (ESMIEPCR4)

This register is dedicated for Group1.

Figure 12-28. ESM Influence $\overline{\text{ERROR}}$ Pin Clear Register 4 (ESMIEPCR4)
[address = FFFF F544h]

31	16
IEPCLR[63:48]	
R/WP-0	
15	0
IEPCLR[47:32]	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

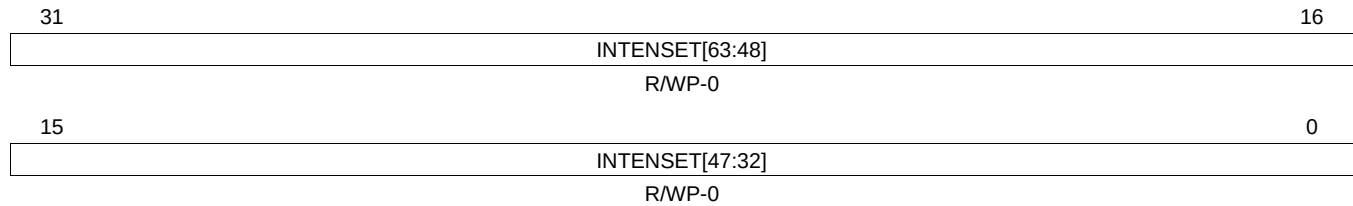
Table 12-20. ESM Influence $\overline{\text{ERROR}}$ Pin Clear Register 4 (ESMIEPCR4) Field Descriptions

Bit	Field	Value	Description
63-32	IEPCLR	0	Clear influence on $\overline{\text{ERROR}}$ pin. Read in User and Privileged mode. Write in Privileged mode only. Read: Failure on channel x has no influence on $\overline{\text{ERROR}}$ pin. Write: Leaves the bit and the corresponding set bit in the ESMIEPSR4 register unchanged.
		1	Read: Failure on channel x has influence on $\overline{\text{ERROR}}$ pin. Write: Disables failure influence on $\overline{\text{ERROR}}$ pin and clears the corresponding set bit in the ESMIEPSR4 register.

12.4.19 ESM Interrupt Enable Set Register 4 (ESMIESR4)

This register is dedicated for Group1.

Figure 12-29. ESM Interrupt Enable Set Register 4 (ESMIESR4)
[address = FFFF F548h]



LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

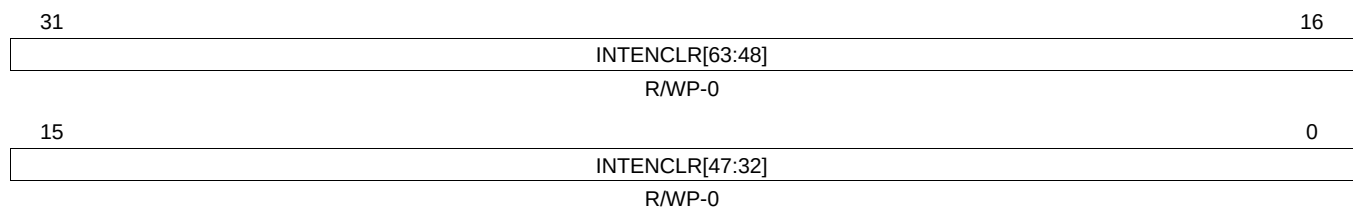
Table 12-21. ESM Interrupt Enable Set Register 4 (ESMIESR4) Field Descriptions

Bit	Field	Value	Description
63-32	INTENSET	0	Set interrupt enable. Read in User and Privileged mode. Write in Privileged mode only. Read: Interrupt is disabled. Write: Leaves the bit and the corresponding clear bit in the ESMIECR4 register unchanged.
		1	Read: Interrupt is enabled. Write: Enables interrupt and sets the corresponding clear bit in the ESMIECR4 register.

12.4.20 ESM Interrupt Enable Clear Register 4 (ESMIECR4)

This register is dedicated for Group1.

Figure 12-30. ESM Interrupt Enable Clear Register 4 (ESMIECR4)
[address = FFFF F54Ch]



LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 12-22. ESM Interrupt Enable Clear Register 4 (ESMIECR4) Field Descriptions

Bit	Field	Value	Description
63-32	INTENCLR	0	Clear interrupt enable. Read in User and Privileged mode. Write in Privileged mode only. Read: Interrupt is disabled. Write: Leaves the bit and the corresponding set bit in the ESMIESR4 register unchanged.
		1	Read: Interrupt is enabled. Write: Disables interrupt and clears the corresponding set bit in the ESMIESR4 register.

12.4.21 ESM Interrupt Level Set Register 4 (ESMILSR4)

This register is dedicated for Group1.

Figure 12-31. ESM Interrupt Level Set Register 4 (ESMILSR4)
[address = FFFF F550h]

31	16
INTLVLSET[63:48]	
R/WP-0	
15	0
INTLVLSET[47:32]	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

Table 12-23. ESM Interrupt Level Set Register 4 (ESMILSR4) Field Descriptions

Bit	Field	Value	Description
63-32	INTLVLSET	0	Set interrupt level. Read in User and Privileged mode. Write in Privileged mode only. Read: Read: Interrupt of channel x is mapped to low-level interrupt line. Write: Leaves the bit and the corresponding clear bit in the ESMILCR4 register unchanged.
		1	Read: Interrupt of channel x is mapped to high-level interrupt line. Write: Maps interrupt of channel x to high-level interrupt line and sets the corresponding clear bit in the ESMILCR4 register.

12.4.22 ESM Interrupt Level Clear Register 4 (ESMILCR4)

This register is dedicated for Group1.

Figure 12-32. ESM Interrupt Level Clear Register 4 (ESMILCR4)
[address = FFFF F554h]

31	16
INTLVLCLR[63:48]	
R/WP-0	
15	0
INTLVLCLR[47:32]	
R/WP-0	

LEGEND: R/W = Read/Write; WP = Write in privileged mode only; -n = value after reset

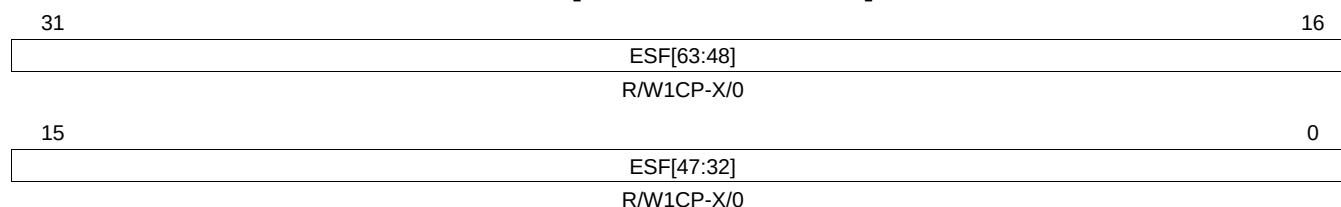
Table 12-24. ESM Interrupt Level Clear Register 4 (ESMILCR4) Field Descriptions

Bit	Field	Value	Description
63-32	INTLVLCLR	0	Clear interrupt level. Read in User and Privileged mode. Write in Privileged mode only. Read: Interrupt of channel x is mapped to low-level interrupt line. Write: Leaves the bit and the corresponding set bit in the ESMILSR4 register unchanged.
		1	Read: Interrupt of channel x is mapped to high-level interrupt line. Write: Maps interrupt of channel x to low-level interrupt line and clears the corresponding set bit in the ESMILSR4 register.

12.4.23 ESM Status Register 4 (ESMSR4)

This register is dedicated for Group1.

Figure 12-33. ESM Status Register 4 (ESMSR4)
[address = FFFF F558h]



LEGEND: R/W = Read/Write; W1CP = Write 1 to clear in privilege mode only; -n = value after reset/ $\overline{\text{PORRST}}$; X = Value unchanged

Table 12-25. ESM Status Register 4 (ESMSR4) Field Descriptions

Bit	Field	Value	Description
63-32	ESF	0	Error Status Flag. Provides status information on a pending error. Read in User and Privileged mode. Write in Privileged mode only. Read: No error occurred; no interrupt is pending. Write: Leaves the bit unchanged.
		1	Read: Error occurred; interrupt is pending. Write: Clears the bit. Note: After $\overline{\text{RST}}$, if one of these flags are set and the corresponding interrupt are enabled, the interrupt service routine will be called.